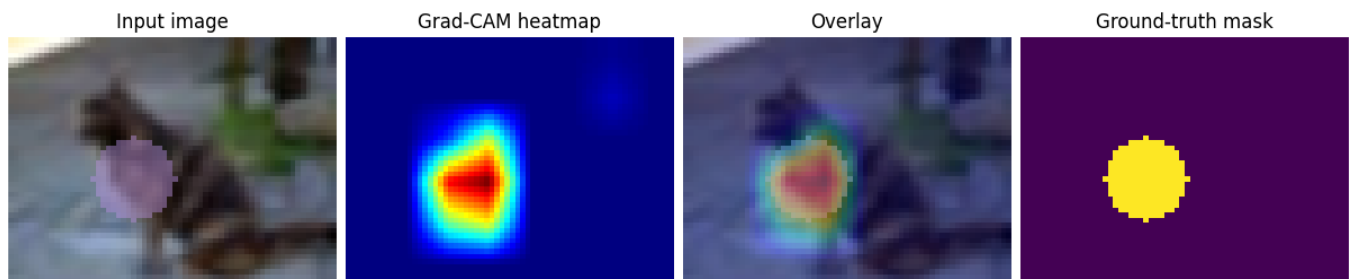


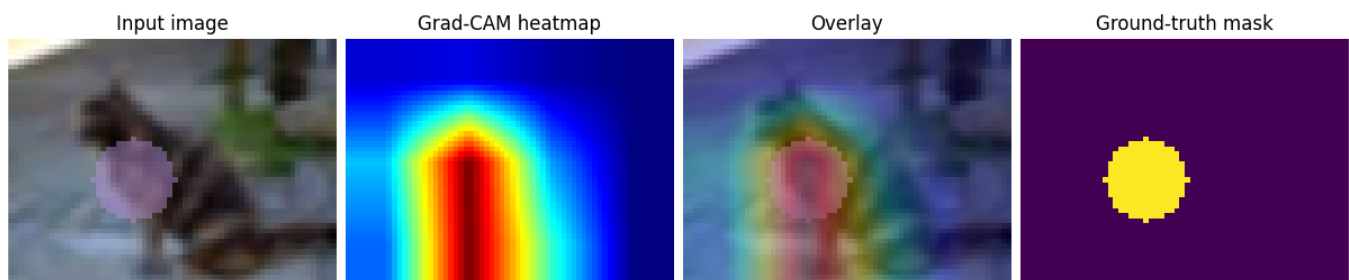
Wojciech Mierzejek 459435 DNN Homework 2

Exemplary visualization of the output of the Grad-Cam

Grad-CAM last layer



Grad-CAM middle layer



SAM Pipeline 1

In the first pipeline I decided to try out providing only one foreground point to the SAM. The point is chosen by calculating average coordinate of 20 points from GradCAM heatmap with highest value.

SAM Pipeline 2

The second pipeline I also base on GradCAM. 5 points from GradCAM with highest value are provided as foreground and 5 points with lowest values are considered background.

Metrics

```
Pipeline 1 (foreground only):  
  Hitrate foreground only: 80.2000%  
  Hitrate foreground and background: 80.2000%  
  Average distance: 4.9514  
  Average intersection over union: 83.8739%  
Pipeline 2 (foreground and background):  
  Hitrate foreground only: 70.3600%  
  Hitrate foreground and background: 85.1200%  
  Average distance: 2.7969  
  Average intersection over union: 73.0895%
```

Discussion

I was surprised with how well the first pipeline performed. I think finding center of mass of GradCAM mask might be improved, perhaps by calculating average of all points above some threshold.